

NETFLIX CATALOG ANALYSIS

POWER BI DASHBOARD REPORT

1925 – 2021

Prepared from the cleaned Netflix titles dataset

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

This project examines the structure, composition, and historical development of the Netflix catalog using 8,807 title records. The analysis covers movies and TV shows released between 1925 and 2021 and added to Netflix between January 2008 and September 2021. Its purpose is to transform a raw title-level dataset into a clean analytical resource and an interactive Power BI dashboard.

The dashboard summarizes catalog size, content type, release-year trends, monthly additions, leading countries, major genre groupings, and title-level details. It provides a compact decision-support view for understanding how the library is distributed and where its strongest concentrations occur.

1.2 Analytical Architecture

- Layer 1 — Data acquisition: the public Netflix titles CSV supplies title, content type, creator, cast, geography, date, rating, duration, genre, and description fields.
- Layer 2 — Data cleaning: Python and Pandas standardize column labels, remove duplicate rows, fill missing categorical values with “Unknown,” fill numeric gaps with medians, and trim whitespace.
- Layer 3 — Analysis: descriptive aggregation measures catalog size, type mix, time patterns, countries, genres, and ratings.
- Layer 4 — Power BI: the cleaned table is presented through KPI cards, filters, charts, and a searchable title-detail table.

1.3 Tools and Technologies

Tool / Technology	Purpose
Python 3	Core language for data preparation and validation
Pandas	CSV ingestion, duplicate removal, missing-value treatment, and aggregation
Matplotlib	Independent validation charts used in this report
Power BI Desktop	Interactive dashboard development and cross-filtering
CSV	Portable source and cleaned-data format
Microsoft Word / PDF	Formal project documentation and distribution

1.4 Project Objectives

- Create a complete, null-free analytical dataset from the original Netflix catalog file.
- Measure the balance between movies and television shows.
- Identify historical growth, seasonal loading patterns, and leading countries.
- Determine the most common genre tags and maturity ratings.
- Deliver an interactive dashboard that supports filtering by date, type, and country.

CHAPTER 2

DATA ACQUISITION

2.1 Data Source and Scope

The source file, `netflix_titles.csv`, contains 8,807 records and 12 columns. Each row represents one Netflix catalog entry. The cleaned version preserves all 8,807 rows and replaces missing text values with “Unknown,” providing a consistent dataset for analysis and dashboard loading.

2.2 Dataset Characteristics

Characteristic	Value
Rows	8,807
Columns	12
Movies	6,131 (69.62%)
TV Shows	2,676 (30.38%)
Release-year range	1925–2021
Date-added range	1 January 2008–25 September 2021
Missing values after cleaning	0

2.3 Raw Data Issues

- Blank director, cast, country, rating, and duration values were present in the source file.
- Text fields could contain leading or trailing whitespace.
- Categorical values required consistent labels for reliable filtering.
- The `date_added` field required parsing before month-based analysis.
- Genre and country fields can contain multiple comma-separated values, which requires care when interpreting category counts.

2.4 Key Data Attributes

Field	Analytical meaning
show_id	Unique catalog identifier
type	Movie or TV Show
title	Program name
director	Credited director or Unknown
cast	Credited performers or Unknown
country	Production country or Unknown
date_added	Date the title entered Netflix
release_year	Original release year
rating	Maturity classification
duration	Minutes for movies or seasons for TV shows
listed_in	One or more genre/category labels
description	Short content synopsis

2.5 Why Cleaning Is Required

Dashboard filters and totals depend on consistent, complete fields. Missing categorical values can disappear from visuals, while whitespace and inconsistent labels create duplicated categories. The cleaning stage therefore protects metric accuracy, ensures every row remains visible, and makes the dataset suitable for repeatable reporting.

CHAPTER 3

DATA PREPROCESSING AND TRANSFORMATION

3.1 Cleaning Workflow

The supplied `datasetcleaning.py` script implements a direct seven-step workflow: load the raw CSV, standardize column names, remove duplicates, fill missing values, trim text, assert that no missing values remain, and export the result.

Step	Operation	Outcome
1	Read CSV into a Pandas DataFrame	Source data available for processing
2	Strip and normalize column names	Consistent snake_case labels
3	Drop exact duplicate rows	Prevents double counting
4	Fill text with Unknown; numeric with median	Complete analytical fields
5	Trim string values	Clean categories and labels
6	Assert zero missing values	Automated quality gate
7	Write cleaned CSV	Reusable Power BI source

3.2 Data Quality Assessment

The cleaned file contains no missing values in any of its 12 columns. The row count of 8,807 matches the expected dashboard total. This validates the central KPI and confirms that null handling did not remove records.

3.3 Important Interpretation Note

Replacing missing values with “Unknown” improves completeness but does not recover the original information. The 831 entries classified under Unknown country should be treated as unassigned records, not as a geographic market.

3.4 Analytical Preparation

- `date_added` is parsed into a date to derive month names and time trends.
- `release_year` is aggregated to show the historical concentration of titles.
- `type` is counted to compare movies and TV shows.
- `country` is ranked by exact field value for the dashboard's leading-market view.
- `listed_in` is split into individual tags for the report's genre analysis; the original dashboard may group full category strings instead.
- `rating` is counted to describe the catalog's maturity profile.

3.5 End-to-End Pipeline

Raw CSV → Python/Pandas cleaning → cleaned CSV → descriptive analysis → Power BI model → interactive dashboard → documented findings.

3.6 Reproducibility

The supplied script contains local Windows file paths. For use on another computer, those paths should be replaced with project-relative paths or configurable input and output arguments. Once corrected, the process can be repeated whenever the source dataset is updated.

CHAPTER 4

ABOUT THIS DASHBOARD

4.1 Dashboard Overview

The Netflix Catalog Dashboard is a single-page interactive report built in Power BI Desktop from the cleaned Netflix titles dataset. It brings together three KPI cards, three slicers, six visuals and a detail table so that the whole catalog can be understood at a glance. This chapter divides the dashboard into its individual parts and explains, for every chart, what it represents, how it was built step by step, and what it shows. Each chart is accompanied by a numbered caption.

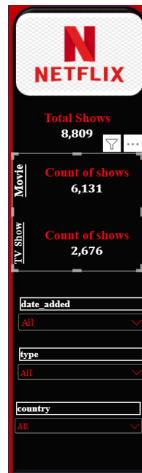


Figure 4.1.1: Left control panel with Netflix logo, KPI cards and slicers.

4.2 How I Built the Dashboard

Step	Action	Result
1	Get Data > Text/CSV: imported the cleaned Netflix dataset into Power BI Desktop	One table named Netflix with 12 columns
2	Power Query: checked data types, set date_added to Date and release_year to Whole Number	Correct sorting and date filtering
3	Created DAX measures such as Total Shows and type-wise counts	Reusable values for KPI cards
4	Set the canvas background to Netflix red with a custom image and added rounded black panels	Branded red/black theme
5	Added the logo, KPI cards and slicers in a left control rail	Simple navigation and filtering
6	Built six visuals and a table, each with a clear underlined title	Complete catalog view on one page
7	Applied data labels, gradient colours and checked cross-filter interactions	Readable and interactive dashboard

4.3 KPI Cards and Slicers

About: The left panel shows the Netflix logo, the **Total Shows** card (8,809) and the **Count of shows** cards for Movies (6,131) and TV Shows (2,676). Below them are three dropdown slicers: **date_added**, **type** and **country**.

How I made it

- Inserted the Netflix logo with **Insert > Image** and placed it at the top-left.
- Added a **Card** visual, dragged show_id into Fields and set the summarisation to **Count** to get Total Shows.
- Added two more cards using the same count and applied a visual-level filter type = Movie and type = TV Show; added rotated text boxes as labels.
- Inserted three **Slicer** visuals with date_added, type and country, and set the style to **Dropdown**.
- Formatted titles in red font on a black background so the slicers match the theme.

What it shows: Users can instantly see the catalog size and the movie/TV split, and the slicers filter every visual on the page at once.

4.4 Count of Shows by Month

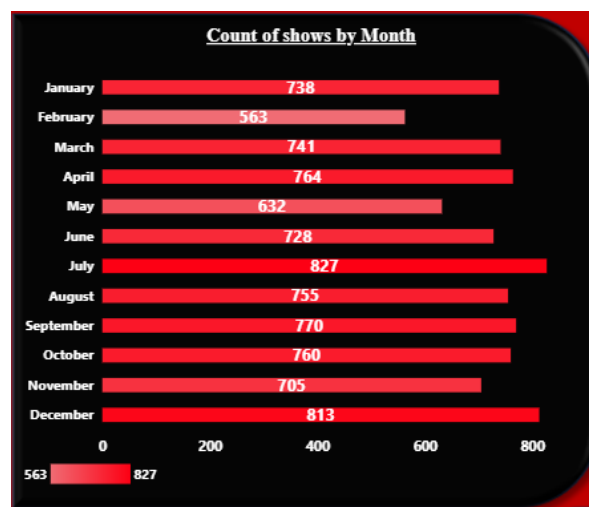


Figure 4.4.1: Count of shows added to Netflix by calendar month (clustered bar chart).

About this chart

A horizontal clustered bar chart that shows how many titles were added to Netflix in each month of the year.

How I made it

- Selected the **Clustered bar chart** visual.
- Placed the month of date_added (from the date hierarchy) on the Y-axis and Count of show_id on the X-axis.
- Sorted the axis by month number so January to December appear in calendar order.
- Turned on **Data labels** inside the bars and applied a red **gradient** conditional colour (563 light to 827 dark).
- Added the title "Count of shows by Month" and rounded the visual border.

What it shows

July has the most additions (827) followed by December (813), while February is the lowest (563).

4.5 Count of Shows by Type

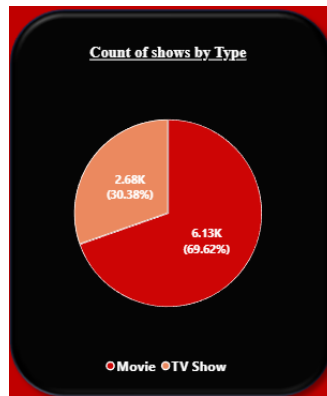


Figure 4.5.1: Share of Movies and TV Shows in the catalog (pie chart).

About this chart

A pie chart that compares the number of Movies with the number of TV Shows.

How I made it

- Selected the **Pie chart** visual.
- Placed type in **Legend** and Count of show_id in **Values**.
- Set slice colours to dark red for Movie and light orange for TV Show.
- Enabled detail labels showing both the value and the percentage of total.
- Moved the legend to the bottom and added the title.

What it shows

Movies make up 6.13K titles (69.62%) and TV Shows 2.68K titles (30.38%), so the catalog is mainly film-led.

4.6 Count of Shows by Year [2000–25]

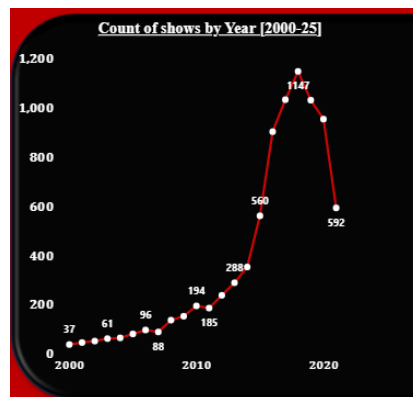


Figure 4.6.1: Count of shows by release year from 2000 onwards (line chart with markers).

About this chart

A line chart showing how many titles in the catalog were released in each year.

How I made it

- Selected the **Line chart** visual.
- Placed release_year on the X-axis (Continuous type) and Count of show_id on the Y-axis.
- Added a visual-level filter so release_year is 2000 or later.
- Turned on **Markers** (white dots) and data labels, and set the line colour to red.
- Added the title "Count of shows by Year [2000-25]".

What it shows

The count rises slowly in the 2000s, grows sharply after 2015, peaks at 1,147 titles in 2018 and falls to 592 in 2021 because the data ends in September 2021.

4.7 Top 5 Countries by Shows

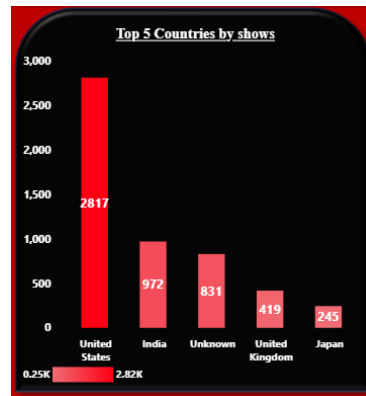


Figure 4.7.1: Top five countries by number of shows (clustered column chart).

About this chart

A vertical column chart showing the five countries that contribute the most titles.

How I made it

- Selected the **Clustered column chart** visual.
- Placed country on the X-axis and Count of show_id on the Y-axis.
- In the Filters pane used **Filter type > Top N**, showing the top 5 items by Count of show_id.
- Sorted the columns in descending order, added data labels and a red gradient colour scale.
- Added the title "Top 5 Countries by shows".

What it shows

The United States leads with 2,817 titles, followed by India (972), Unknown (831), the United Kingdom (419) and Japan (245).

4.8 Title Detail Table

show_id	title	type	release_year
1	Dick Johnson Is Dead	Movie	2020
10	The Starling	Movie	2021
100	On the Verge	TV Show	2021
1000	Stowaway	Movie	2021
1001	Wild Dog	Movie	2020
1002	Olubiri	Movie	2015
1003	Tell Me When	Movie	2021
1004	Zero	TV Show	2021
1005	Lazy's Koala World	TV Show	2021
1006	Keymon and Nani in Space Adventure	Movie	2013
1007	Motu Patlu Dino Invasion	Movie	2018
1008	Motu Patlu in	Movie	2017

Figure 4.8.1: Title-level detail table with show ID, title, type and release year.

About this chart

A table that lists individual records behind the summary charts.

How I made it

- Selected the **Table** visual.
- Added the columns show_id, title, type and release_year.
- Formatted the header with a red background and white bold text, and added gridlines.
- Applied alternating red and white row text so each record is easy to read.
- Kept a scroll bar so users can browse the full list.

What it shows

The table changes with every slicer or chart selection, making it easy to find exactly which titles make up a number.

4.9 Total Shows by Genre

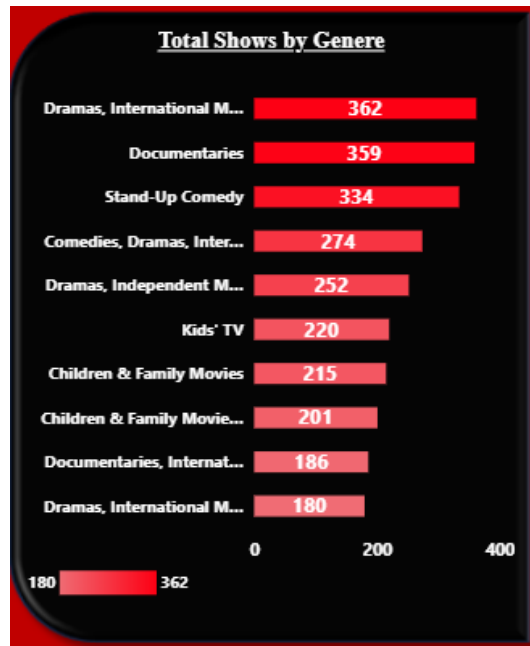


Figure 4.9.1: Top ten genre groups by total shows (clustered bar chart).

About this chart

A horizontal bar chart that ranks the most common genre groups from the listed_in column.

How I made it

- Selected the **Clustered bar chart** visual.
- Placed listed_in on the Y-axis and Count of show_id on the X-axis.
- Applied a **Top N** filter to keep the top 10 genre groups.
- Sorted in descending order, added data labels and a red gradient (180 to 362).
- Added the title "Total Shows by Genre".

What it shows

"Dramas, International Movies" is the largest group (362), followed by Documentaries (359) and Stand-Up Comedy (334).

4.10 Formatting and Design Choices

- **Theme:** Netflix red background with black rounded panels to match the Netflix brand.
- **Colours:** red gradients show higher values with stronger colour; light orange marks TV Shows.
- **Titles:** every chart has an underlined white title for quick identification.
- **Labels:** data labels on all charts so exact values are visible without hovering.
- **Interactivity:** all visuals cross-filter each other and respond to the three slicers.

4.11 Summary of Dashboard Charts

The table below summarises every element of the dashboard, the Power BI visual used, the fields placed on it and its caption number in this chapter.

Element	Visual type	Fields used	Figure
Total Shows KPI	Card	Count of show_id	4.1.1
Movie / TV Show KPIs	Card (filtered)	Count of show_id, type	4.1.1
Slicers	Slicer (dropdown)	date_added, type, country	4.1.1
Count of shows by Month	Clustered bar chart	Month of date_added, Count of show_id	4.4.1
Count of shows by Type	Pie chart	type, Count of show_id	4.5.1
Count of shows by Year	Line chart	release_year, Count of show_id	4.6.1
Top 5 Countries	Clustered column chart	country (Top N = 5), Count of show_id	4.7.1
Title detail table	Table	show_id, title, type, release_year	4.8.1
Total Shows by Genre	Clustered bar chart	listed_in (Top N = 10), Count of show_id	4.9.1

4.12 Conclusion of the Chapter

By dividing the dashboard into its KPI cards, slicers, six charts and a detail table, this chapter shows how each part was built in Power BI and what it tells the reader. Together, these visuals answer the main catalog questions — how large the catalog is, how it splits by type, when titles were released and added, which countries and genres lead, and which records sit behind each number — on a single interactive page.

CHAPTER 5

POWER BI DASHBOARD DEVELOPMENT

5.1 Objective of the Dashboard

The Power BI dashboard consolidates the cleaned dataset into a single interactive report page. It is designed to answer six practical questions: how large the catalog is, how it splits by type, when titles were released or added, which countries contribute most, which genre groupings dominate, and which records sit behind the summary metrics.

5.2 Dashboard Layout

The page uses Netflix red and black branding. A left control rail presents the logo, KPI cards, and slicers. The main area contains monthly and yearly trends, a type-distribution chart, a country ranking, genre ranking, and a title-detail table.

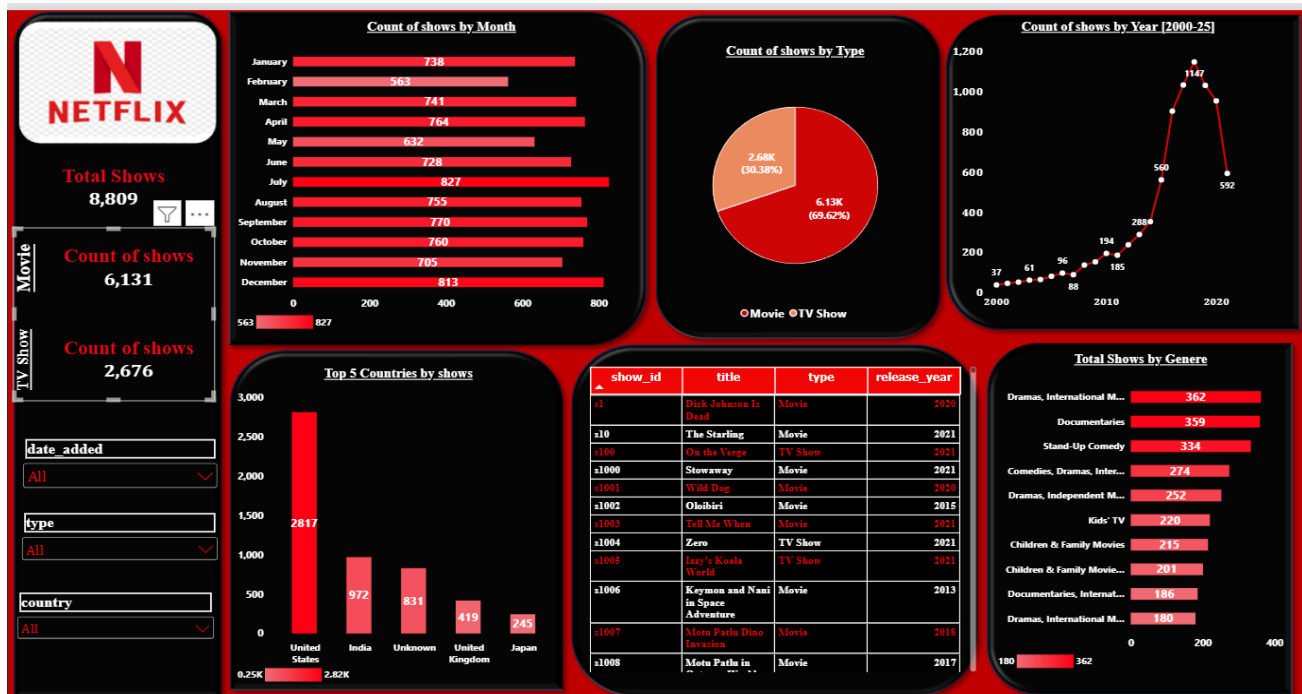


Figure 5.2.1: Complete Netflix Power BI dashboard supplied with the project.

5.3 Interactivity

- Date-added slicer narrows the page to selected loading dates.
- Type slicer switches between Movie and TV Show results.
- Country slicer focuses all visuals on a selected geographic value.
- Chart selections cross-filter the remaining visuals and title table.

5.4 Visual-by-Visual Interpretation

Visual	Purpose	Observed result
Total Shows KPI	Catalog scale	8,809 shown in screenshot; cleaned source validates 8,807
Movie KPI	Movie count	6,131
TV Show KPI	TV show count	2,676
Count by Month	Seasonal loading pattern	July leads with 827; February has 563
Count by Type	Catalog mix	Movies 69.62%; TV Shows 30.38%
Count by Year	Release history	Peak at 1,147 in 2018
Top 5 Countries	Geographic concentration	United States leads; India second
Genre bars	Category concentration	Dashboard ranks combined listed_in strings
Detail table	Record inspection	Show ID, title, type, and release year

5.5 Validation Note

The dashboard screenshot displays Total Shows as 8,809, while both the uploaded raw and cleaned CSV files contain 8,807 data rows. The Movie and TV Show cards sum to 8,807. The two-record difference therefore appears to be a stale KPI, an alternate imported source, or a counting configuration issue. The Total Shows measure should be recalculated as COUNTROWS of the cleaned table or DISTINCTCOUNT of show_id.

5.6 Recommended Measures

Measure	Definition
Total Shows	COUNTROWS(Netflix) or DISTINCTCOUNT(Netflix[show_id])
Movie Count	CALCULATE([Total Shows], Netflix[type] = "Movie")
TV Show Count	CALCULATE([Total Shows], Netflix[type] = "TV Show")
Movie Share	DIVIDE([Movie Count], [Total Shows])
TV Show Share	DIVIDE([TV Show Count], [Total Shows])

5.7 Dashboard Strengths

- All major catalog questions are visible on one page.
- The left-side slicers provide direct and familiar filtering.
- The detail table connects summary patterns to individual titles.
- Red/black styling creates immediate visual association with Netflix.

5.8 Areas for Improvement

- Correct the Total Shows discrepancy so every KPI reconciles to 8,807.
- Use a dedicated calendar table for reliable year and month ordering.
- Split multi-value country and genre fields into bridge tables for precise contribution analysis.
- Increase label contrast and simplify heavy borders to improve readability.
- Add a rating chart or tooltip because maturity classification is analytically important.

CHAPTER 6

FINDINGS, RESULTS AND OUTCOMES

6.1 Key Findings from the Dashboard

- Scale: the cleaned dataset contains 8,807 Netflix titles across 12 attributes.
- Type mix: movies account for 6,131 titles (69.62%), while TV shows account for 2,676 (30.38%).
- Release trend: the catalog is heavily concentrated in recent titles, peaking at 1,147 releases from 2018.
- Monthly pattern: July has the most additions at 827; February has the fewest at 563.
- Geography: the United States leads with 2,818 exact country entries, followed by India with 972.
- Metadata quality: 831 titles have Unknown country values, making this the third-largest country category.
- Genres: International Movies, Dramas, and Comedies are the most frequent individual tags.
- Ratings: TV-MA and TV-14 jointly account for 5,367 titles, or 60.9% of the catalog.

6.2 Interpretation

The historical Netflix library represented by this dataset is film-heavy, recent-release-oriented, and globally diverse, but strongly concentrated in United States and Indian content. International and drama categories dominate the genre profile, while the rating mix emphasizes mature and teen audiences. The dashboard is effective for descriptive catalog analysis, provided its total KPI is reconciled with the source data.

6.3 Outcomes of the Project

- A complete cleaned dataset with no missing values.
- A repeatable Python cleaning workflow.
- A branded, interactive Power BI dashboard.
- Validated descriptive findings for catalog type, time, country, genre, and rating.
- A documented KPI discrepancy and a clear correction path.

6.4 Limitations

- The dataset ends on 25 September 2021 and does not describe the current Netflix catalog.
- Missing values were labeled Unknown rather than recovered from an authoritative source.
- Comma-separated country and genre fields complicate exact category attribution.
- The dataset describes catalog availability, not viewing behavior, subscribers, revenue, or title performance.
- The supplied PBIX binary could not be independently inspected here; dashboard interpretation is based on the supplied screenshot and datasets.

6.5 Future Scope

- Automate dataset refresh and preserve a refresh timestamp on the dashboard.
- Normalize countries, genres, cast members, and directors into separate analytical tables.
- Add movie-duration and TV-season distributions.
- Compare catalog additions with original release year to measure acquisition lag.
- Introduce drill-through pages for countries, genres, directors, and individual titles.
- Add data-quality KPIs for Unknown fields and duplicate detection.

REFERENCES

- Netflix Titles Dataset. Uploaded files: netflix_titles.csv and cleaned_netflix_dataset.csv.
- Python cleaning script. Uploaded file: datasetcleaning.py.
- Netflix Power BI Dashboard. Uploaded files: Netfilx_Dashboard.pbix and dashboard_screenshot.png.
- Pandas documentation: data cleaning, missing values, datetime parsing, and aggregation.
- Microsoft Power BI documentation: data modeling, DAX measures, slicers, and visual interactions.

APPENDIX: CORE VERIFIED METRICS

Metric	Verified value
Total records	8,807
Movies	6,131
TV Shows	2,676
Movie share	69.62%
TV Show share	30.38%
Peak release year	2018 (1,147 titles)
Peak addition month	July (827 titles)
Leading country value	United States (2,818 titles)
Leading genre tag	International Movies (2,752 occurrences)
Leading rating	TV-MA (3,207 titles)